S-E: local admission, causal transitions and an allocation gate

15 October 2026 — analytical qualification, no traffic experiment

Status. S-E is the user-selected first qualification direction. The common trial constitutive law and the S-E-P/S-E-W allocations below are proposals, not approved task equations. Stop at **S-E-A2**, the free-A allocation interpretation. The whole three-part review and Step 3 remain unfinished. No source theorem is transferred to this node.

1. Retained interface and a concrete trial state-to-work law

H1 supplies terminal entrance histories F_j with the accepted common constant advice $v \in [v_-, v_+]$. The terminal has length b, initially empty histories, and no internal lateral source. Put

$$C(v) = \frac{wk_j v}{w+v}, \quad P_j(t) = F_j(t - b/v), \quad q_j = P_j - Y_j, \quad N_j = F_j - Y_j, \quad j = A, E, B. \tag{1}$$

Let $\lambda_j = \dot{P}_j$ denote eligible arrival rates. B retains $Y_B = P_B$ on its independent adequate-service domain. For A/E, let $B(t)$ denote available shared work, $0 \le B \le R \le C(v_-)$. Here B is an input to an instantaneous audit, not an externally supplied future receiving trace. Its eventual law must be autonomous and causal.

Use mode $m_j = F$ for free/unqueued admission and $m_j = Q$ for a selected queued boundary trace. Initially $q_j = 0, m_j = F$. An active queue has demand $C(v)$; an empty free approach offers λ_j. The proposed equilibrium trace identification is

$$u_j = v \ (m_j = F), \qquad u_j = U(g_j) := \frac{wg_j}{wk_j - g_j} \ (m_j = Q), \quad 0 \le g_j \le C(v), \quad U(C(v)) = v. \tag{2}$$

An absent free stream's formal v is not a measured speed. Equation (2) is an upstream boundary trace, not mean lane speed. Identifying that trace with a representative insertion/recovery mark remains an adaptation.

One concrete trial constitutive family is

$$c_A = 1, \qquad c_E = 1 + \delta + \beta \frac{g_A}{C(v)} \frac{[u_A - u_E]_+}{v}, \qquad \delta, \beta \ge 0, \qquad g_A + c_E g_E \le B. \tag{3}$$

c_j is work per admitted mass. δ represents immediate maneuver work; it may be zero. The additional one-sided term represents harder entry when A is faster, weighted by A's actual passing flow. No absent A stream produces a mismatch penalty. This is not an empirically identified formula or a universal sign theorem. The finite late work and its separation from post-entry recovery need physical qualification. $c_A = 1$ normalizes ordinary current work; it does not forbid a justified A recovery influence.

2. What is unique inside a selected regime

Retain the proposed fair-work allocator as a *comparison template*: for demands d_A, d_E, costs $1, c_E$ and budget B, choose minimal $z \geq 0$ such that

$$\min(d_A, z) + \min(c_E d_E, z) = \min(B, d_A + c_E d_E), \quad g_A = \min(d_A, z), \quad g_E = \min(d_E, z/c_E). \tag{4}$$

This is equal work with unused-work redistribution, not Carey's demand-proportional R2 or a literature-derived fairness theorem.

When E is free, $u_E = v$ and $c_E = c_0 := 1 + \delta$ for either A trace. Thus (4) has the previously proved unique fixed-input allocation. When E is queued, its work offer reaches B. Under (4),

$$a := g_A = \min(d_A, B/2), \qquad x := B - a, \qquad H_E(g) := g\left[c_0 + \beta \frac{a}{C(v)} \frac{[u_A - U(g)]_+}{v}\right] = x. \tag{5}$$

For a free-A-priority branch with A still served freely, instead take $a = \lambda_A \leq B$ and $x = B - a$ in (5). In both cases u_A is fixed by that selected A mode and a.

A conservative sufficient condition for a unique feasible root is

$$\eta := 1 + \delta - \beta(1 + v_+/w) > 0. \tag{6}$$

Indeed $U'(g) = w^2 k_j/(wk_j - g)^2$ and $gU'(g) \leq v(1 + v/w)$ on $[0, C(v)]$. On the active positive-part branch,

$$H'_E(g) = c_0 + \beta \frac{a}{C(v)v}\{u_A - U(g) - gU'(g)\} \ \geq \eta; \tag{7}$$

off it $H'_E = c_0$. At the kink the continuous function remains strictly increasing. Also $H_E(0) = 0$ and $H_E(C(v)) = c_0 C(v) \geq B \geq x$. Consequently (5) has exactly one root in $[0, C(v)]$, with

$$\frac{x}{c_0 + \beta} \leq g_E \leq \frac{x}{c_0}, \qquad 0 \leq g_A + g_E \leq B. \tag{8}$$

Condition (6) is a mathematical sufficient domain, not an approved parameter restriction on real traffic. It need not be necessary. No parameter is fitted or chosen for a positive control result.

Limit check. With both streams queued under (4), $a = B/2$ and $g_E \leq a$. If $\delta = 0$, the unique solution is $g_E = a$ with matched traces and $c_E = 1$. If traces match with $\delta > 0$, the mismatch term vanishes but the declared maneuver work remains. Recovery may still reduce B. Comparable speeds do not by themselves prove high total service.

3. Two allocations and the causal transition rules

S-E-W, immediate cooperation. Apply (4) in every mode. Free A does not have an additional protection rule. Its service can be reduced while all E arrivals pass. Both queued streams retain positive service when $B > 0$.

S-E-P, free-A priority then queued cooperation. While A is empty and its free offer fits B, put

$$g_A = \lambda_A, \quad g_E = \begin{cases} \min\{\lambda_E, (B - \lambda_A)/c_0\}, & m_E = F, \\ H_E^{-1}(B - \lambda_A), & m_E = Q. \end{cases} \tag{9}$$

If A cannot fit B, its strict shortfall triggers its queued mode. Whenever A is queued, use (4)–(5), including a free E offer if applicable. This proposed regime adaptation is not strict A priority in all states: with both queues nonempty, A does not consume all available service.

For either allocation the proposed S-E transition semantics are:

1. Test admission in the current free modes. If every empty approach's eligible offer is met, keep those approaches free. An alternative queued root is not an activation condition.

2. A strict free-mode shortfall selects the affected queued trace; recompute the coupled node and require nonnegative right-hand queue growth for a newly active empty queue. In the strict onset examples below, this condition holds. No external seed or numerical queue epsilon is used.

3. While $q_j > 0$, evolve $\dot{q}_j = \lambda_j - g_j$ and $\dot{Y}_j = g_j$. At a true zero hit, test free admission again. Never serve unavailable mass. Equality alone does not trigger a new queued branch.

4. Retain the common recovery history across queue clearance. At load jumps use physical right-hand data. Simultaneous events and grazing need qualification in the complete coupled law.

These rules reject the old arbitrary-birth diagnostic: its free work demand is $7/10 < 4/5$, so no activation is available under the constant local inputs. That is a local exclusion of the specific delayed births, not a global well-posedness theorem. The fixed-regime inverse bound (6) and strict-event checks do not prove uniqueness through every coupled switching event.

The distinction between (9) and (4) selects which stream first waits and which departure speeds feed later recovery. S-E direction approval does not select that allocation, an exact equal-work convention, or (3).

4. A reachable H1 front exposes the allocation difference

Reuse the accepted joined H1/T1 algebraic family, in units with its reference speed and density equal to one:

$$\begin{gathered} v \in [1,2], \quad w = 2, \quad k_j = 3, \quad e_0 = 7/10, \quad a_0 = b_0 = 1/4, \quad \gamma = 2, \\ R = 39/20, \quad R_B = 1, \quad b = \ell > 0, \quad L_p = \ell + h \log 2, \quad L = L_p + b, \quad h > 0. \end{gathered} \tag{10}$$

The existing exact H1 preparation gives front ending density $e_f(2) = 125/352$. At the first eligible front for $v = 2$,

$$C = 3, \quad \lambda_E = 125/176, \quad \lambda_A = 1047/880, \quad \lambda_A + \lambda_E = 19/10 < R. \tag{11}$$

The terminal and history are empty, hence $B = R$ for an initially neutral continuous recovery account. For this exact static comparison only, take $\delta = \beta = 1/10$ in (3); these values are not selected task parameters. Then $\eta = 9/10 > 0$ and

$$c_0\lambda_E = 25/32 < R/2, \qquad \lambda_A + c_0\lambda_E - R = 37/1760 > 0. \tag{12}$$

S-E-W. All E arrivals fit their fair-work share. The causal onset is A waiting, with

$$g_E = 125/176, \quad g_A = 187/160, \qquad \dot{q}_A = 37/1760 > 0, \quad \dot{q}_E = 0. \tag{13}$$

The queued A trace becomes $374/773$, while free E retains $v = 2$ and $c_E = c_0$. This is a compatible right-hand regime for the trial law.

S-E-P. Free A first passes its full offer. The free-E test gives

$$g_E^F = 669/968 < 125/176, \qquad \lambda_E - g_E^F = 37/1936 > 0. \tag{14}$$

Thus the declared maneuver-work shortfall activates queued E. Its rate solves

$$g\left[\frac{11}{10} + \frac{1}{10}\frac{1047/880}{3}\left(1 - \frac{g}{6-g}\right)\right] = \frac{669}{880}, \quad 5189g^2 - 33432g + 20070 = 0. \tag{15}$$

Only

$$g_* = \frac{16716 - 183\sqrt{5234}}{5189}, \quad \frac{2230}{3343} < g_* < \frac{669}{968} < \frac{125}{176} \tag{16}$$

is feasible. A remains free; E has a strictly positive queue-growth margin. The strict inequalities persist locally under continuous histories and right-continuous arrival data. This is an H1-reachable local onset check, not a time-stepped traffic trajectory or complete cohort result.

5. A simpler demand-proportional graft fails the mode test

Carey's R2 and demand-based merge models are important simpler rivals. Their native discretization and a properly invariant LTM junction cannot be identified with the following naive graft. Test, only for this audit,

$$g_A = \sigma d_A, \quad g_E = \sigma d_E, \quad \sigma = \min\left\{1, \frac{B}{d_A + c_E d_E}\right\}, \quad d_j = \begin{cases} \lambda_j, & m_j = F, \\ C, & m_j = Q. \end{cases} \tag{B1}$$

At the same initially empty H1 front (11), there is no compatible strict right-hand mode under these binary demand/trace rules:

- Both free: $\sigma = 3432/3469 < 1$. Both arrivals are restricted, so neither empty approach can stay free with zero waiting derivative.
- Only A queued: E is free, so $c_E = c_0$ and $g_A = 936/605 > 1047/880 = \lambda_A$. A's newly active empty queue would have a negative derivative. E's free offer is also restricted.
- Only E queued: $1.1 \le c_E \le 1.2$ gives

  $$g_E = \frac{3B}{\lambda_A + 3c_E} \ge \frac{1716}{1405} > \frac{125}{176} = \lambda_E. \tag{B2}$$

  The newly active empty E queue would have a negative derivative.
- Both queued: equal mass demands give equal actual flows and hence equal congested traces. Thus $c_E = c_0$ and

  $$g_A = g_E = \frac{B}{1 + c_0} = \frac{13}{14} > \frac{125}{176} = \lambda_E. \tag{B3}$$

  Again E's newborn queue cannot have that negative derivative.

The change from an empty demand to the capacity offer immediately alters the proportional allocation. Merely saying "activate the queue on a shortfall" does not complete (B1). A numerical method might alternate between branches, but its step must not select a scientific flux law.

This excludes the specific binary-demand graft, not all demand-based merging, native R2 or every possible generalized solution. An explicit interior-state/selection law would be an additional model. The primary LTM source itself distinguishes invariant and non-invariant fair merges and gives a stationary counterexample for the latter in section 6. Our transient work example is a separate deduction; its proof is not attributed to that paper. Neither S-E-P nor S-E-W is declared generally invariant or globally well posed on the strength of this test.

6. Joined implications, proof limits and the stopping gate

Physical interface. In (10), $M = 6\ell/5 < 3\ell = k_j b$ and all H1 entrance rates fit $C(v)$ across $[1, 2]$. For any eventual admissible departures, the retained finite-terminal vacancy bound gives $V_j(t) \geq k_j b - M > 0$ after b/w; the early-time bound follows as before from $\dot{F}_j \leq C(v) < wk_j$. The handoff protection does not depend on which allocation is selected. No point-buffer substitution is made. Conservation and complete local TST retain

$$N_j = F_j - Y_j, \qquad J(v) = \frac{M(L - \ell/2)}{v} + \int_0^\infty (q_A + q_E + q_B)dt. \tag{17}$$

Completed mass leaves once; service memory is not physical stock.

Recovery is not closed by the admission check. The two onsets place slow boundary traces in different streams. E-only recovery would be neutral at the initial S-E-W onset; that does not prove its whole shared account neutral. A's actual slow passage may have a justified continuing influence. Neither deleting that influence nor charging every queued A departure independently is permitted without qualification. Both allocations retain one account, not independent ceilings per past departure.

For orientation, the earlier isolated kernel may be marked by a justified actual insertion speed u, rather than a fixed E speed. If only a separately justified E source were present, a candidate $D(\theta; u)$ would satisfy

$$p_E(t) = m_{\text{ref}}^{-1} \int_0^t D(t - s; u_E(s))dY_E(s), \quad 0 \leq D(\theta; u) \leq D(\theta; 0), \quad \int_0^\infty p_E(t)dt \leq \frac{MW(0)}{m_{\text{ref}}}. \tag{18}$$

This is a conditional extension bound, not an adopted E-only meter, overlap rule or full M2 proof. If an admitted full account establishes $\int (R - B)dt \leq \Lambda$ and $c_E \leq c_{\max} = 1 + \delta + \beta$, either work-conserving allocation clears after last eligibility $t_* = L/v$ by the previously derived sufficient bound

$$t_* + \frac{c_{\max}(q_A(t_*) + q_E(t_*)) + \Lambda}{R}. \tag{19}$$

No positive service floor or transferred M1 uniqueness proof is used.

Evidence disposition. Proved: unique fixed-regime roots under (6), specific causal-onset consistency, rejection of the old spontaneous-birth diagnostic, and a reachable local H1 allocation contrast. Retained: conditional storage/accounting/clearance interfaces. Not proved: whole-law uniqueness through simultaneous/grazing events, the actual joint recovery law, complete-cohort MP6, observed merge credibility, an action split or gradient usefulness. Conditional scalar reconstruction remains a strong rival.

Gate S-E-A2 (Type 2 G3). Choose whether free A first claims its offer until reduced shared service makes it queue (S-E-P), or cooperation may slow A from the first conflict (S-E-W). Provisional first qualification: S-E-P, retaining S-E-W as a serious comparator. Neither allocation is selected. This does not reopen S-E versus S-V, H1, the primary demand domain or the local objective. The exact trial cost/trace identification remains a hypothesis. Stop before selecting the allocation and dependent shared recovery.